Surrogate Losses and Binary Classification

Assignment 6 — Experimental Project

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Abstract

We study how three convex surrogate losses affect the optimization and predictive performance of linear binary classifiers. We implement independently a hinge-loss linear support vector machine (SVM), a logistic-regression classifier, and a least-squares classifier, each trained from scratch by full-batch gradient-based optimization with L2 regularization. We evaluate the models on linearly separable synthetic data, nonlinear/noisy synthetic data, and the UCI QSAR Biodegradation dataset. The same stratified splits and training-only standardization are used for every loss. All classifiers attain zero test classification error on the separable data. On the non-separable data, logistic loss yields the lowest test error (19.3%); on QSAR, hinge loss yields the lowest (13.4%). Training trajectories show that decreasing a surrogate objective does not guarantee a decrease in the zero–one classification error at every update. An additional regularization sweep illustrates how changes in coefficient magnitude need not translate into monotone changes in predictive accuracy. These findings are descriptive for the stated datasets and splits, not population-level rankings of the losses.

1 Problem and objective

For binary labels $y_i \in \{-1, +1\}$ and feature vectors $x_i \in \mathbb{R}^d$, all three algorithms use the same affine score and decision rule:

$$f_{w,b}(x) = w^\top x + b, \qquad \hat{y}(x) = \begin{cases} +1 & f_{w,b}(x) \geq 0, \\ -1 & f_{w,b}(x) < 0. \end{cases} \tag{1}$$

The empirical classification error is $\widehat{R}_{01} = \frac{1}{n}\sum_{i=1}^{n} \mathbf{1}\{\hat{y}(x_i) \neq y_i\}$. Directly minimizing this discontinuous objective is difficult. Convex surrogates replace the zero–one penalty by a function of the signed margin $m_i = y_i f_{w,b}(x_i)$, giving a tractable empirical-risk minimization problem [1]. Our objective is empirical: holding the hypothesis class, train/test partitions, and preprocessing fixed, we test how the *choice of surrogate* changes the learned classifier, error trajectories, and decision boundary.

2 Classifiers and implementation

2.1 Three independent loss implementations

Each classifier is written in a separate Python module: `src/svm.py`, `src/logistic_regression.py`, and `src/least_squares.py`. No prebuilt machine-learning estimator performs training. The three objectives differ in their loss $\phi(m)$:

$$\textbf{Hinge (linear SVM)}: \quad \phi_{\mathrm{h}}(m) = \max\{0, 1-m\}, \qquad \partial\phi_{\mathrm{h}}(m) = \begin{cases} -1 & m < 1, \\ 0 & m \geq 1; \end{cases} \tag{2}$$

$$\textbf{Logistic}: \quad \phi_{\mathrm{l}}(m) = \log(1 + e^{-m}), \qquad \phi_{\mathrm{l}}'(m) = -\frac{1}{1+e^{m}}; \tag{3}$$

$$\textbf{Squared}: \quad \phi_{\mathrm{s}}(m) = \frac{1}{2}(1-m)^2, \qquad \phi_{\mathrm{s}}'(m) = m - 1. \tag{4}$$

For the hinge loss we select the valid subgradient zero at the nondifferentiable point $m = 1$. Logistic loss uses numerically stable `logaddexp` calculations. With $y_i \in \{-1, +1\}$, squared margin loss equals $\frac{1}{2}(y_i - f(x_i))^2$, so it defines a least-squares classifier rather than probability-based logistic regression.

2.2 Optimization

For a selected loss, all classifiers minimize the L2-regularized empirical objective

$$J_\lambda(w, b) = \frac{1}{n}\sum_{i=1}^{n} \phi\big(y_i(w^\top x_i + b)\big) + \frac{\lambda}{2}\|w\|_2^2. \tag{5}$$

The bias b is not regularized. Setting $g_i = y_i \phi'(m_i)$ (using a subgradient for hinge), the full-batch derivatives and updates are

$$\nabla_w J_\lambda = \frac{1}{n}\sum_i g_i x_i + \lambda w, \quad \partial_b J_\lambda = \frac{1}{n}\sum_i g_i, \quad (w^+, b^+) = (w, b) - \eta(\nabla_w J_\lambda, \partial_b J_\lambda). \tag{6}$$

All weights and the bias start at zero. At each epoch the optimizer computes a full-training-set (sub)gradient, proposes a step of size $\eta = 0.6$, and halves the step as needed until the training objective does not increase (allowing a numerical tolerance of 10^{-12}). At most 35 trial step sizes are considered, after which the old parameters are restored if no candidate is accepted. We use $\lambda = 0.01$ and 180 epochs in the main comparison. For a nonsmooth hinge objective, nonincreasing objective values do not by themselves establish convergence to the exact optimum.

2.3 Evaluation measures

We record each epoch's *unregularized* mean surrogate risk $\widehat{R}_\phi = n^{-1} \sum_i \phi(m_i)$, the regularized training objective J_λ, zero–one error, and weight norm. When test data are supplied, test surrogate risk and test classification error are recorded without using them to choose any parameter updates. Importantly, surrogate values from *different* losses have different scales: a logistic loss of 0.3 is not intrinsically better or worse than a squared loss of 0.2. Cross-loss predictive comparison therefore uses the common zero–one error metric.

3 Datasets and evaluation protocol

3.1 Datasets

Three binary datasets are considered. Both synthetic datasets contain 600 two-dimensional observations drawn using a fixed seed, so that the resulting decision boundaries can be displayed directly.

- **Linearly separable:** draw $x \sim U([-3, 3]^2)$, retain only points for which $|x_1 + 0.7x_2 - 0.1| > 0.65$, and assign $y = \text{sign}(x_1 + 0.7x_2 - 0.1)$. This creates a margin gap around a linear boundary.
- **Non-separable:** draw $x \sim U([-3, 3]^2)$, assign labels from $\text{sign}(x_1 + 0.85 \sin(2.2x_2) - 0.15)$, and randomly flip 17% of labels, selecting rows without replacement. The generating rule is nonlinear and deliberately noisy.
- **QSAR biodegradation:** use the authentic UCI `biodeg.csv` containing 41 molecular descriptors and the label RB/NRB [2]. Of 1,055 original rows, three exact duplicates are removed, leaving 1,052 unique observations; RB is encoded as $+1$ and NRB as -1. The original descriptor names are assigned to the headerless file.

Table 1: Dataset sizes and class distributions after data preparation.

Dataset	Features	n	Class -1	Class $+1$	Train/Test
Linearly separable	2	600	318	282	450/150
Non-separable	2	600	292	308	450/150
QSAR biodegradation	41	1052	698	354	790/262

3.2 Shared split and preprocessing

Each dataset is divided once into a stratified 75% training and 25% test partition using seed 42. On QSAR, exact duplicates are removed *before* splitting, preventing identical rows from appearing in both partitions. Feature means and standard deviations are estimated only on training observations and then applied unchanged to train and test sets; near-zero standard deviations are replaced by one. Each loss is trained on precisely the same scaled data. We use a single split for a controlled descriptive comparison, not repeated cross-validation.

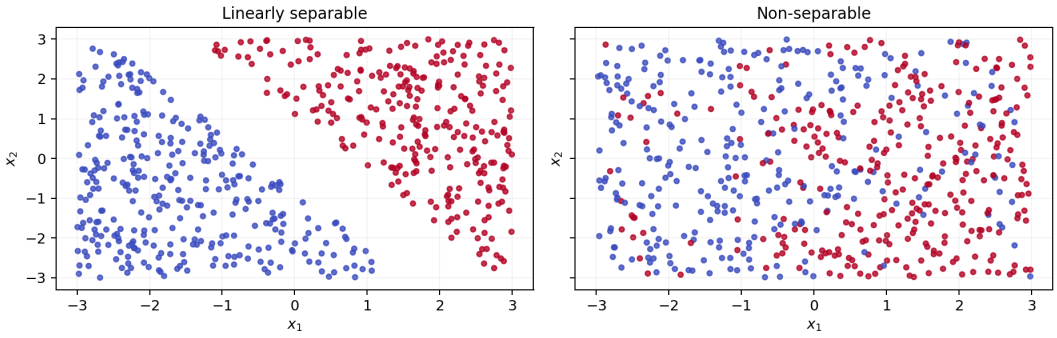


Figure 1: Two synthetic learning problems. Color denotes the true class; the non-separable problem combines a nonlinear label rule with randomly flipped labels.

4 Experimental results

4.1 Overall classification performance

Table 2: Final outcomes at epoch 180 ($\lambda = 0.01$): zero–one classification error and mean unregularized surrogate risk. Risk magnitudes should be compared across epochs *within* a loss, not across different loss functions. Best test error within each dataset is in bold.

		0–1 error (%)		Surrogate risk	
Dataset	Loss	Train	Test	Train	Test
Linearly separable	Hinge	0.00	**0.00**	0.0098	0.0152
	Logistic	0.00	**0.00**	0.0725	0.0715
	Squared	0.00	**0.00**	0.0892	0.0899
Non-separable	Hinge	26.00	20.67	0.6231	0.4967
	Logistic	26.22	**19.33**	0.5527	0.4810
	Squared	26.44	20.00	0.3724	0.3104
QSAR biodegradation	Hinge	12.28	**13.36**	0.3146	0.3302
	Logistic	12.53	15.65	0.3101	0.3474
	Squared	13.80	16.03	0.2255	0.2693

On the separable data, all three models reach zero classification error on both partitions, despite differing objective functions. In particular, zero–one error cannot distinguish the models once their decision boundaries correctly separate every observed point. Their mean training surrogate risks remain positive: approximately 0.0098 (hinge), 0.0725 (logistic), and 0.0892 (squared), reflecting distinct margin penalties rather than residual misclassifications.

On the non-separable dataset, all three classifiers retain substantial irreducible error under the chosen linear hypothesis class. Logistic loss has the lowest observed test error (19.33%), but its margin over squared (20.00%) and hinge (20.67%) is small. Test error is below training error for all three models on this particular split; this is a sampling outcome and should not be mistaken for evidence of superior generalization.

On QSAR, hinge loss achieves the lowest observed test error (13.36%), followed by logistic (15.65%) and squared (16.03%). The class distribution is imbalanced (698 NRB and 354 RB observations); a majority-class predictor would make approximately 33.6% errors on the held-out set. All three fitted linear models outperform that simple baseline on this split. The result is consistent with meaningful predictive signal in the 41 descriptors, but the single split cannot establish statistically significant superiority of hinge loss.

4.2 Learning trajectories and margin geometry

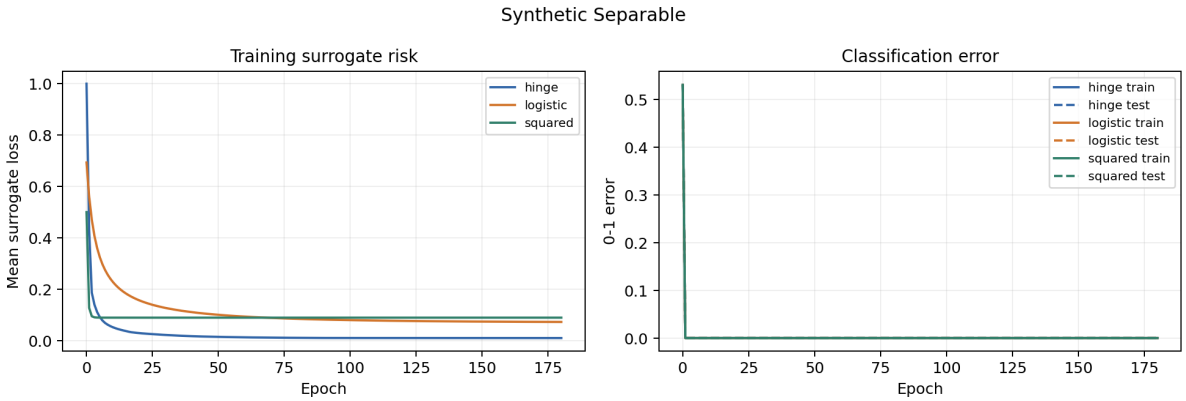


Figure 2: Separable data: training surrogate risk (left) and train/test zero–one error (right) across epochs. Different loss scales should not be compared vertically.

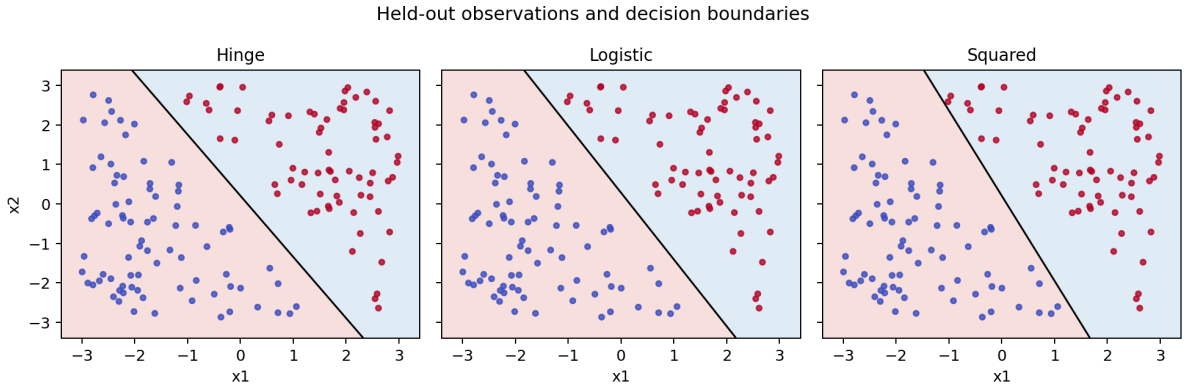


Figure 3: Fitted linear decision boundaries on the separable dataset (black), shown with held-out observations. All three methods classify the held-out observations correctly.

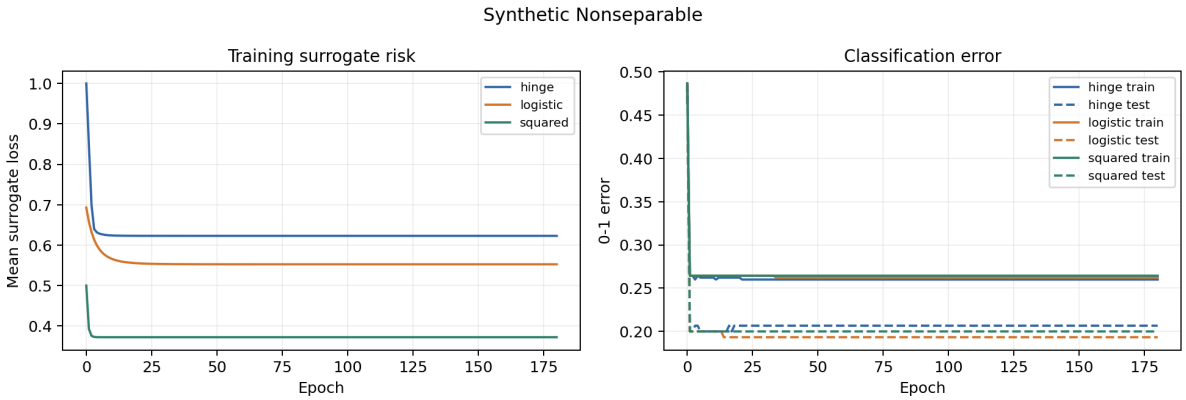


Figure 4: Non-separable data: mean training surrogate loss and train/test classification error by epoch. Error trajectories quickly develop plateaus while surrogate penalties continue to change.

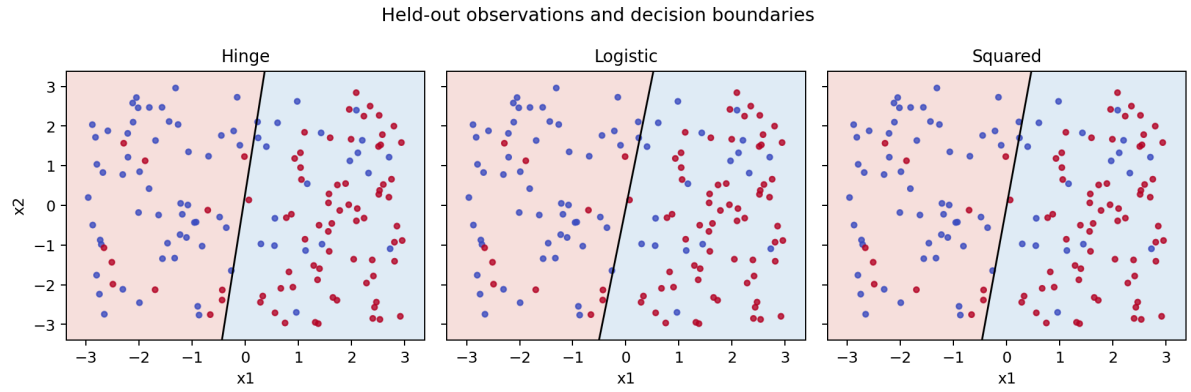


Figure 5: Learned boundaries on nonlinear and label-noisy synthetic data. Each classifier is constrained to a straight-line decision boundary, despite the nonlinear label-generating mechanism.

The separable learning curves reach zero classification error rapidly, while their respective surrogate risks decrease by different amounts and at different rates (Figure 2). The learned boundaries are similar but not identical (Figure 3). On the non-separable data, the classifier family cannot represent the nonlinear label-generating boundary exactly. Figure 4 shows how the classifiers adapt their linear scores; Figure 5 shows the resulting geometric compromises.

4.3 Relationship between surrogate risk and classification error

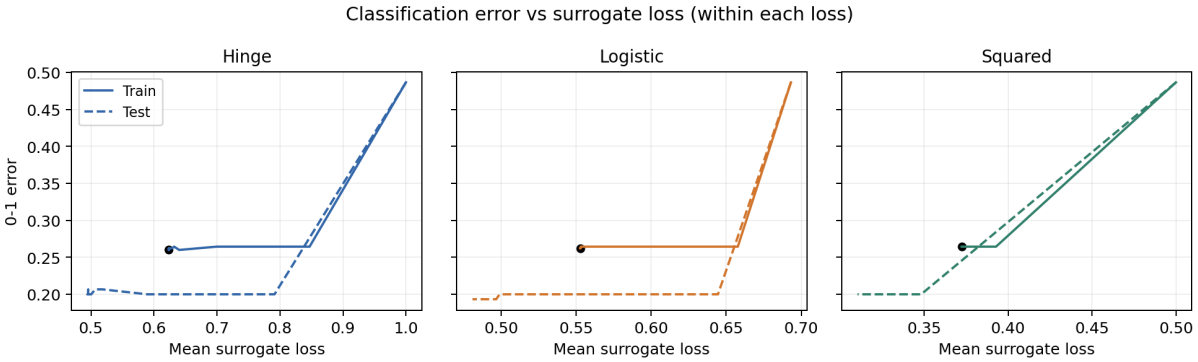


Figure 6: Non-separable data: zero–one error against each loss's *own* surrogate risk along training. Separate horizontal axes are necessary because loss values have different interpretations.

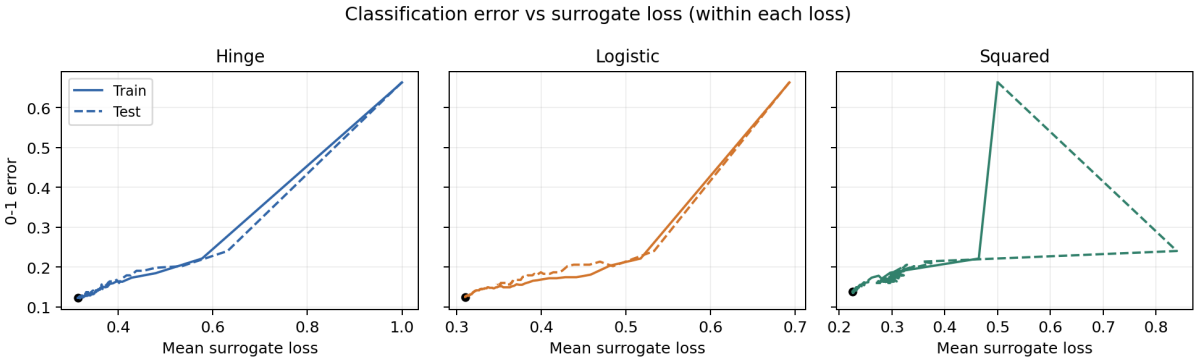


Figure 7: QSAR: classification error versus surrogate risk, for training (solid) and test (dashed) trajectories. Lower surrogate loss does not force the error to fall monotonically.

Surrogate risks respond to score and margin changes even when no predicted label flips. Zero–one error, in contrast, changes only when a score crosses the decision threshold. The distinction is visible in the flat stretches of Figures 6–7. The separation is not just a plateau effect: on QSAR, hinge training risk decreases in 24 epochs where training error nevertheless increases, while squared-loss training risk decreases in 64 such epochs. Thus an optimization improvement need not yield an immediate classification improvement. These counts describe

individual epoch-to-epoch changes under the implemented numerical protocol; they are not estimates of a population-risk relationship.

4.4 QSAR training behavior

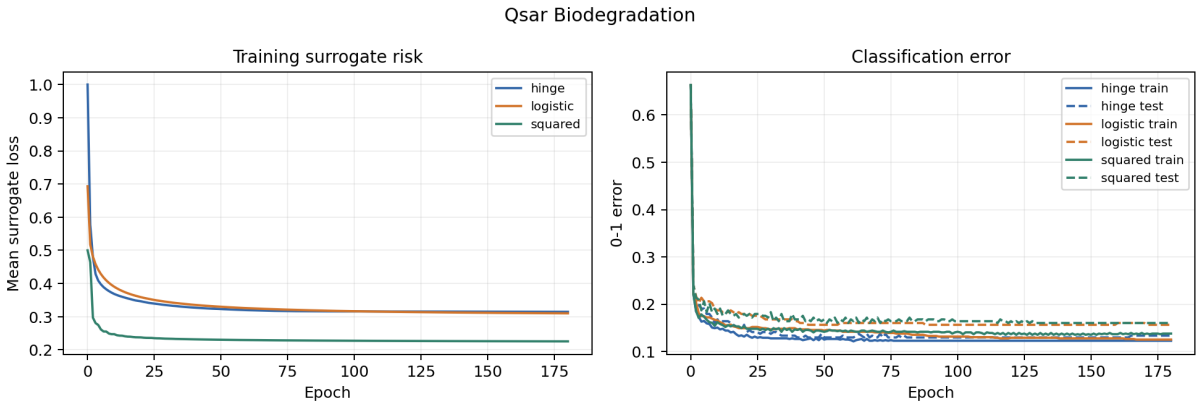


Figure 8: QSAR optimization trajectories: training surrogate loss (left) and train/test classification error (right). All methods use the same standardized features and data split.

The QSAR curves show steady progress in the training objectives but more irregular zero–one errors. For example, the final hinge test error is 0.1336 and logistic test error is 0.1565; this ranking concerns classification decisions, not the absolute numerical values of their different surrogate losses. Monitoring test curves is useful diagnostically, but our update rule uses only the training objective, and no model selection is performed with the test partition.

5 Extension: L2 regularization sensitivity

We vary $\lambda \in \{0, 0.001, 0.01, 0.1, 1\}$ on the non-separable synthetic dataset, holding the split, standardization, learning rate and 120-epoch budget fixed. The larger penalty generally reduces $\|w\|_2$ (Figure 9), but the test error is not monotone in λ. For example, hinge test error is 20.67% at $\lambda = 0.01$, 20.00% at $\lambda = 0.1$ and 25.33% at $\lambda = 1$. Logistic test error is 19.33% at $\lambda = 0.01$ and 20.67% at $\lambda = 0.1$. Hence smaller coefficient norm does not, by itself, imply a smaller held-out error. The sweep is a sensitivity study, not an unbiased search for a best hyperparameter; the same held-out set is plotted for all configurations.

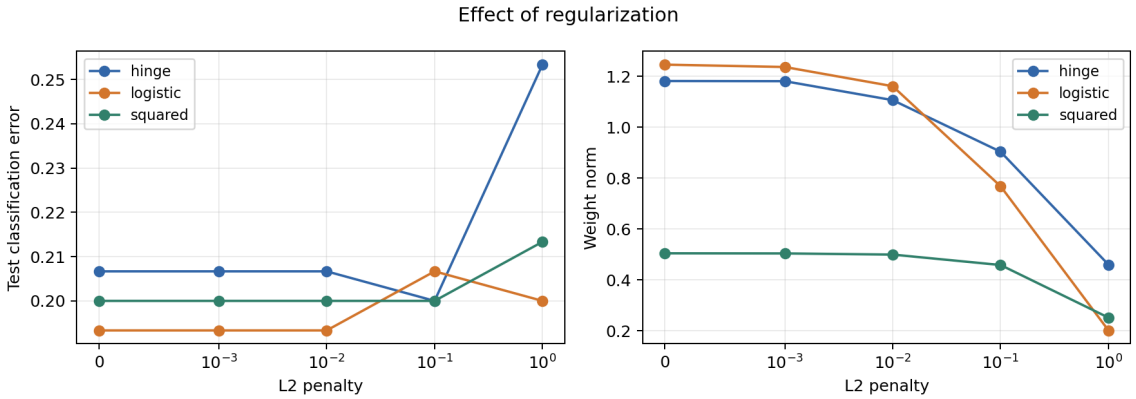


Figure 9: Non-separable dataset: held-out classification error (left) and weight norm (right) for five L2 strengths. Each point is a separate full-batch training run of 120 epochs.

6 Discussion, limitations and conclusion

This controlled experiment demonstrates that a common linear hypothesis class can be trained with three distinct convex losses and produce different optimization trajectories, boundaries and held-out error rates. Hinge assigns

no loss to observations with margin at least one; logistic penalizes every finite margin, decreasing smoothly with margin; squared loss drives signed scores toward one and also penalizes scores with margins greater than one. These different margin preferences explain why equal classification error need not correspond to equal surrogate risk or identical parameters. The empirical results agree with the basic motivation for surrogate learning [1], without testing or proving the paper's theoretical risk bounds.

Several limitations restrict interpretation. First, a single stratified split per dataset provides no uncertainty estimate or significance test; repeated splits or cross-validation would be needed to assess robustness of rankings. Second, the study compares only affine predictors, so its non-separable errors reflect model-class limitations as well as the choice of loss. Third, the synthetic noise process is designed for controlled illustration and does not emulate QSAR chemistry. Fourth, 180 training epochs (120 for the regularization sweep) constitute a fixed computational budget; monotonicity under backtracking is not a certificate of exact numerical convergence. Finally, QSAR class imbalance makes class-conditional performance and calibrated probability assessment worthwhile follow-up analyses, but these are not evaluated here.

In summary, all three losses successfully train linear predictors from scratch. Their observed ranking depends on the dataset: indistinguishable test zero–one error on the separable data, a slight logistic advantage on the nonlinear/noisy synthetic data, and a hinge advantage on QSAR. Most importantly, the study directly illustrates that *optimization progress in a surrogate objective and improvement in classification error are not the same quantity.*

Reproducibility and code organization

The repository contains `src/svm.py`, `src/logistic_regression.py`, `src/least_squares.py` (independent from-scratch classifiers), `src/data.py` (data generation and train-only scaling), `src/experiments.py` (consistent evaluation), and `src/plots.py` (figures). The four numbered notebooks generate/inspect data, run the loss comparisons, and study regularization. With the included `Data/` files, notebook 03 reproduces the primary metrics and notebook 04 reproduces the regularization study. Experiment seeds, epochs, learning rate and L2 values are specified in Section 3 and the notebook cells.

References

[1] P. L. Bartlett, M. I. Jordan, and J. D. McAuliffe, "Convexity, Classification, and Risk Bounds," *Journal of the American Statistical Association*, 101(473), 138–156, 2006. DOI: 10.1198/016214505000000907.

[2] K. Mansouri, T. Ringsted, D. Ballabio, R. Todeschini, and V. Consonni, "QSAR biodegradation" [dataset], *UCI Machine Learning Repository*, 2013. DOI: 10.24432/C5H60M.

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